### 2.2.1 Acceleration and velocity change

Average acceleration is final velocity minus initial velocity, divided by elapsed time. A change in speed or direction can cause acceleration. Negative acceleration means acceleration in the chosen negative direction, not necessarily slowing down. Use a consistent sign convention for all velocities before calculating their difference.

$$a_{\text{av}} = \frac{v - u}{\Delta t}$$

u, v: initial and final velocity (m s$^{-1}$); Δt: time (s); $a_{\text{av}}$: average acceleration (m s$^{-2}$). Signed values along one chosen axis; equals instantaneous acceleration when constant.

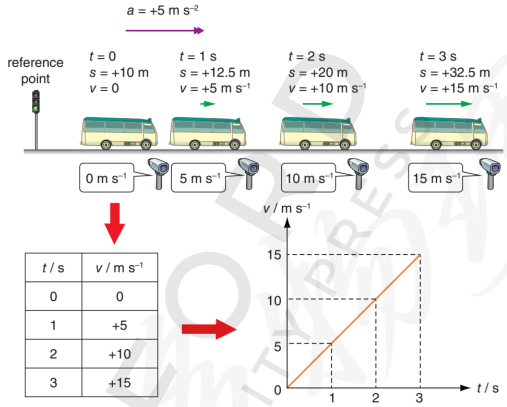



*Fig 2.2i Plotting the v–t graph of a car in uniformly accelerated motion.*

Equal increases in velocity in equal times produce a straight v–t graph. Its slope is acceleration; its signed area gives displacement.

### 2.2.2 Using velocity and acceleration to identify speeding up

In one dimension, an object speeds up when velocity and acceleration have the same direction and slows down when they oppose each other. Check both signs: negative velocity with negative acceleration means increasing speed in the negative direction. Zero instantaneous velocity does not imply zero acceleration, as at the top of a vertical throw.

### 2.2.3 Slope of a position-time graph

The slope of a position-time graph gives velocity. A secant between two points gives average velocity over the interval. A tangent gives instantaneous velocity. The sign indicates direction, and a horizontal line means unchanged position. The vertical coordinate is position, so a higher point does not itself mean a greater speed.

$$v = \frac{ds}{dt}$$

Velocity is the tangent slope of a displacement–time graph. Over an interval Δs/Δt gives average velocity. s: m; t: s; v: m $s^{-1}$.

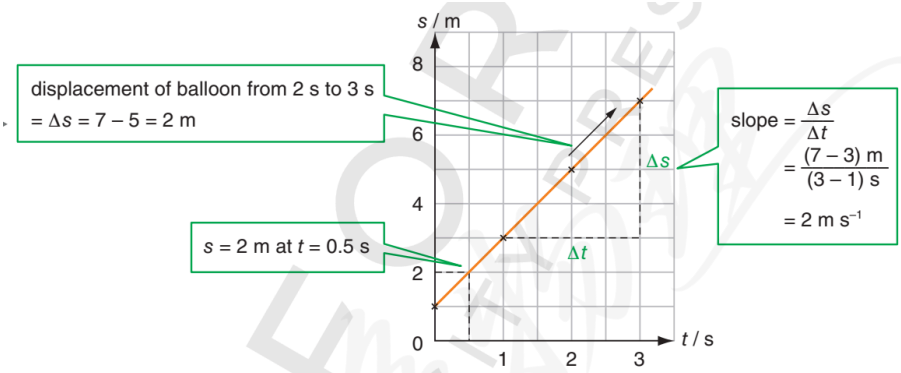



***Fig 2.2b*** *How position (s), displacement (Δs) and average velocity (Δs/Δt) are found from an s–t graph.*

The slope is Δs/Δt: the marked rise of 4 m in 2 s gives 2 m $s^{-1}$. Position is read from the vertical axis; displacement is a change in position.

### 2.2.4 Slope of a velocity-time graph

The slope of a velocity-time graph gives acceleration. A straight line has constant acceleration. Upward or downward slope indicates the sign of acceleration, but whether the object speeds up also depends on velocity. A horizontal line means constant velocity. The object is at rest only if that line lies on the time axis.

$$a = \frac{\mathrm{d}v}{\mathrm{d}t}$$

Acceleration is the tangent slope of a velocity–time graph. Over an interval Δv/Δt gives average acceleration. v: m $s^{-1}$; t: s; a: m $s^{-2}$.

### Area under a velocity-time graph

Signed area between a velocity-time graph and the time axis gives displacement: areas above the axis are positive and those below are negative. To find distance travelled, add the magnitudes of the separate areas instead. Triangles and trapezia can be calculated geometrically.

$$\Delta s = \int_{t_1}^{t_2} v \, \mathrm{d}t$$

Displacement equals the signed area under a velocity–time graph. Area below the time axis is negative; distance uses the sum of absolute areas. Integral notation represents this area rule.

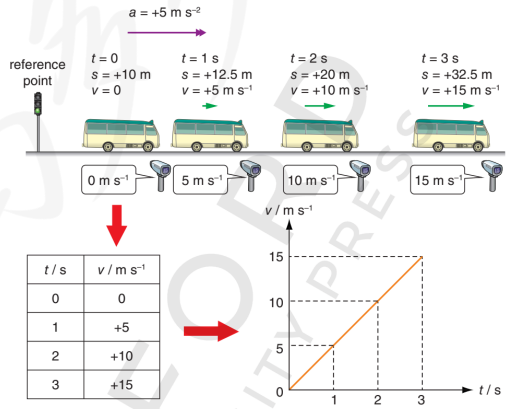



*Fig 2.2i* *Plotting the v–t graph of a car in uniformly accelerated motion.*

Equal increases in velocity in equal times produce a straight v–t graph. Its slope is acceleration; its signed area gives displacement.

### 2.2.5 Connections between motion graphs

The slope of position-time gives velocity, and the slope of velocity-time gives acceleration. Conversely, signed area under acceleration-time gives the change in velocity, and signed area under velocity-time gives displacement. Areas give changes only. Add the initial velocity or position when reconstructing actual values and retain the appropriate signs.

$$s \xrightarrow{\text{slope}} v \xrightarrow{\text{slope}} a$$

$$\Delta s = \int_{t_1}^{t_2} v \, \mathrm{d}t$$

$$\Delta v = \int_{t_1}^{t_2} a \, \mathrm{d}t$$

For graphs against time: slopes give velocity then acceleration; signed areas give changes in velocity or displacement. Initial values are needed to recover v or s.

$$s\text{–}t \text{ graph} \underset{\text{area}}{\overset{\text{slope}}{\rightleftarrows}} v\text{–}t \text{ graph} \xrightarrow{\text{slope}} a\text{–}t \text{ graph}$$

Read slopes from displacement to velocity to acceleration. Read signed areas in reverse to find changes, adding the appropriate initial value.

### 2.2.6 Equations of uniformly accelerated motion

These equations apply to straight-line motion with constant acceleration. Choose a positive direction, identify initial and final velocities, displacement and time, then select an equation without unnecessary unknowns. Displacement is not distance travelled. If acceleration differs between stages, solve each stage separately rather than treating the whole motion as uniformly accelerated.

$$v = u + at$$

$$s = \frac{u + v}{2}t$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

Constant acceleration along one axis. u, v: velocities (m s$^{-1}$); a: acceleration (m s$^{-2}$); s: displacement (m); t: time (s). Choose a positive direction consistently.

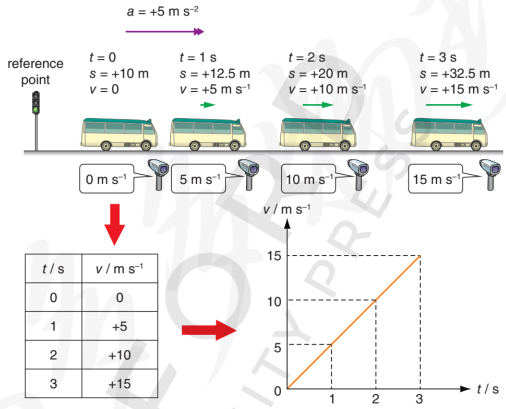



*Fig 2.2i* Plotting the v–t graph of a car in uniformly accelerated motion.

Equal increases in velocity in equal times produce a straight v–t graph. Its slope is acceleration; its signed area gives displacement.

### 2.2.7 Free fall and vertical projection

Near Earth's surface, neglecting air resistance, objects of different masses have the same downward gravitational acceleration, approximately 9.81 m s$^{-2}$. A rising object still accelerates downwards. At its highest point velocity is momentarily zero but acceleration is not. With upwards positive, acceleration is negative throughout the flight.

$$g \approx 9.81\ \text{m s}^{-2}$$

Near Earth's surface; negligible air resistance. Acceleration is vertically downward, including during upward motion.